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Hannah Scholz

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Master's Thesis Mathematics

Advisor: Dr. Michael Rothgang

Second Advisor: Prof. Dr. Floris van Doorn

Mathematisches Institut

Mathematisch-Naturwissenschaftliche Fakultät der

Rheinischen Friedrich-Wilhelms-Universität Bonn

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1 The mathematics of collar neighborhoods

1.1 Prerequisites

Lemma 1.1.1 (Smooth Urysohn's lemma, in mathlib I think)**.** *Let M be a $\mathcal{C}^k$-manifold and A and B two disjoint closed sets in M. Then there is a C^k-function $f\colon M \to [0,1]$ such that f is 0 on A and 1 on B.*

Remark 1.1.2. Again, this requires a definition of manifold that makes it metrizable.

1.2 Basics about collar neighbourhoods

Definition 1.2.1. A *topological manifold* is a second-countable Hausdorff space that is locally Euclidean.

Remark 1.2.2. Apparently, there are some subtleties with collars of non-metrizable manifolds but I assume we don't care. (At least with the above definition our manifolds are always metrizable.)

Definition 1.2.3. Let M and N be two C^k-manifolds. Let $f\colon M \to N$.

(i) f is a C^k-diffeomorphism if it is bijective and both f and its inverse functions are k-times continuously differentiable.

(ii) f is a C^k-embedding if it is a C^k-diffeomorphism onto its image.

Definition 1.2.4 (Due to [Con71])**.** Let M be a $\mathcal{C}^k$-manifold (with boundary) and let $B \subseteq M$ be closed.

(i) B is *locally collared* if it is covered by sets $U \subseteq B$ which are open in B such that there are $\mathcal{C}^k$-embeddings $\varphi_U\colon \overline{U} \times [0,1] \to M$ with

  a) $\varphi_U^{-1}(B) = \overline{U} \times \{0\}$.

  b) For all $x \in \overline{U}$, $\varphi_U(x,0) = x$.

  c) $\varphi_U(\overline{U} \times [0,1])$ is closed.

  d) $\varphi_U(U \times [0,1)$ is open.

(ii) B is *collared* if B already fulfils the conditions given above. In this case we call the image of $B \times [0,1)$ under the corresponding map $\varphi_B\colon B \times [0,1] \to M$ a *collar neighbourhood* of B.

Remark 1.2.5. This definition is a lot more restrictive than other definition which don't consider the closure or the closed interval. I think if this one doesn't work, I can just take the other one.

Remark 1.2.6. For $k = 0$ we can even extend this definition to topological spaces instead of just defining it for topological manifolds.

Theorem 1.2.7 (Due to [Con71])**.** *Let X be a Hausdorff space and $B \subseteq X$ be a closed subset. If B is locally collared in X, then it is collared.*

Proof. Let $U_1, \ldots, U_n$ be a cover of B that witnesses that B is locally collared. Let $\varphi_i \colon \overline{U_i} \times [0,1] \to X$ be the corresponding embeddings. B is normal because it is compact and Hausdorff. By the shrinking lemma (already in mathlib) there is a new open cover $V_1, \ldots, V_n$ of B such that $\overline{V_i} \subseteq U_i$ for all $1 \leq i \leq n$. Define

$$X^+ \coloneqq (X \amalg B \times [-1,0])/x \sim (x,0).$$

Clearly, X^+ is collared. So it suffices to show that there is a homeomorphism $G \colon X \to X^+$ s.t. $G(x) = (x,-1)$ for all $x \in B$. Let $\Phi_i \colon \overline{U_i} \times [-1,1] \to X^+$ be defined by

$$\Phi_i(x) = \begin{cases} \phi_i(x) & \text{if } x \in \overline{U_i} \times [0,1] \\ x & \text{if } x \in \overline{U_i} \times [-1,0]. \end{cases}$$

We now inductively define maps $f_i \colon B \to [0,1]$ and embeddings $g_i \colon X \to X^+$ for all $0 \leq i \leq n$ such that

(i) $f_i(x) = -1$ if $x \in \bigcup_{j \leq i} \overline{V_j}$

(ii) $g_i(x) = (x, f_i(x))$ if $x \in B$

(iii) $g_i(X) = X \cup \{(x,t) \mid t \geq f_i(x)\}$

Then g_n will be the desired homeomorphism G from above. Set f_0 to be constant 0. Then we can choose g_0 to be the inclusion and trivially $g_0(X) = X = X \cup X \times \{0\} = X \cup \{(x,t) \mid t \geq 0\}$.

Assume that g_{i-1} has been defined. Using Urysohn's Lemma, choose a continuous function $\lambda_i \colon \overline{U_i} \to [0,1]$ such that it is 0 on $\overline{U_i} \setminus U_i$ and 1 on $\overline{V_i}$. Given $x \in B$ set $a(x) \coloneqq f_{i-1}(x)$ and $b(x) \coloneqq (1 - \lambda_i(x)) f_{i-1}(x) + \lambda_i(x)(-1)$. Now let

$$s(x, \_) \colon [a(x), 1] \to [b(x), 1], \quad t \mapsto \frac{b(x) - 1}{a(x) - 1}(t-1) + 1.$$

Now set

$$\psi_i \colon \Phi_i^{-1}(g_{i-1}(X)) \to \overline{U_i} \times [-1,1], \quad (x,t) \mapsto (x, s(x,t)).$$

This allows us to define

$$\Psi_i \colon g_{i-1}(X) \to X^+, \quad x \mapsto \begin{cases} \Phi_i(\psi_i(\Phi_i^{-1}(x))) & \text{if } x \in g_i(X) \cap \Phi_i(\overline{U_i} \times [-1,1]) \\ x & \text{otherwise} \end{cases}$$

Claim 1: Ψ_i is continuous.

Proof. It suffices to show that ψ_i is continuous. We need to show that $s \colon B \times [a(x), 1] \to [b(x), 1]$ is continuous.

Ahhh this is ugly. I somehow would prefer to just get going with this in Lean. □

□

I want to try to separate out some parts here:

Definition 1.2.8. Let X be a $(n+1)$-dimensional $\mathcal{C}^k$-manifold and $B \subseteq X$ a closed subset with some nice properties (n-dimensional). We define the topological space

$$X_B^+ \coloneqq (X \amalg B \times [-1,0])/x \sim (x,0).$$

Lemma 1.2.9. *Let X be an n-dimensional $\mathcal{C}^k$-manifold and $B \subseteq X$ a closed subset. Then X_B^+ is an n-dimensional $\mathcal{C}^k$-manifold.*

Proof. We first need to show that X_B^+ is a Hausdorff space. I am not sure if this is best done manually or if there is some cool lemma like: a pushout along closed embeddings preserves Hausdorff spaces. (Is this true?)

Then we need to show that X_B^+ is second countable. Again, there should be some general lemma about closed embeddings and pushouts.

Now we need to pick an atlas for X_B^+. Let $\mathcal{A}_X$ be a $\mathcal{C}^k$-atlas for X, $\mathcal{A}_B$ be a $\mathcal{C}^k$-atlas for B and $\mathcal{A}_{[-1,0]}$ be a $\mathcal{C}^k$-atlas for $[-1,0]$. We define

$$\begin{aligned}\mathcal{A} \coloneqq &\{(U,\varphi) \in \mathcal{A}_X \mid U \cap B = \varnothing\}\\ &\cup \{(U \times I, \varphi \times \psi) \mid (U,\varphi) \in \mathcal{A}_B, (I,\psi) \in \mathcal{A}_{[-1,0]}, 0 \notin I\}\\ &\cup \{(U \cup ((U \cap B) \times I), \varphi_I^+) \mid (U,\varphi) \in \mathcal{A}, (I,\psi) \in \mathcal{A}_{[-1,0]}, 0 \in I\}\end{aligned}$$

where we define φ_I^+ as follows. Let (U,φ) and (I,ψ) as above. Set

$$\varphi_I^+ \colon U \cup ((U \cap B) \times I) \to \mathbb{R}^n, x \mapsto \begin{cases} \varphi(x) & x \in U \\ (\varphi(y), z) & x = (y,z) \in (U \cap B) \times I \end{cases}$$

For $x \in B$ we have $\varphi(x)_I^+ = \varphi(x) = (\varphi(x), 0) = \varphi_I^+(x,0)$, so φ_I^+ is continuous.

It seems to me that this might not be $\mathcal{C}^k$ at all. I don't know of you can even glue this without basically already having the vector field available and then one might as well do the other proof. Maybe a local vector field is enough? Not sure :(

Now you would need to verify that this is a $\mathcal{C}^k$ atlas. Possibly using a version of the manifold chart lemma from Lee. I will only try to show that the transition maps are $\mathcal{C}^k$. The rest should be easier. The only interesting intersections are those of two charts of the form $(U \cup ((U \cap B) \times I), \varphi_I^+)$ and $(V, \cup((V \cap B) \times J), \psi_J^+)$. We need to check that the transition map is $\mathcal{C}^k$ on $U \cap V \cap B$. This is clear for all combinations of partial derivatives that do not include the n-th one. □

Lemma 1.2.10. *Let X be an n-dimensional $\mathcal{C}^k$-manifold and $B \subseteq X$ a closed subset. Then X_B^+ is the pushout in the category of $\mathcal{C}^k$-manifolds.*

Remark 1.2.11. It seems that this is not at all trivial and in fact much of the issue with gluing manually along boundaries. I am not sure if we can even guarantee the above lemma. The above lemma would hopefully mean that one never has to look at the explicit construction again.

Definition 1.2.12. Let X be a Hausdorff space and $B \subseteq X$ a closed subset. Let $U \subseteq B$ be an open subset together with a

We would like the following theorem:

Theorem 1.2.13. *Let M be a $\mathcal{C}^k$-manifold (with boundary) and $B \subseteq X$ a closed subset. If B is locally collared in M, then it is collared.*

Then we will need the following lemma:

Lemma 1.2.14. *Let M be an n-dimensional $\mathcal{C}^k$-manifold with boundary. Then ∂M is locally collared.*

Proof. Let $p \in \partial M$. Then p is contained in a regular half-coordinate ball U, i.e. there is a coordinate half-ball $U' \supseteq \overline{U}$ and a coordinate map $\psi \colon U' \to \mathbb{H}^n$ such that for some $r' > r > 0$ we have $\psi(U) = B_r(0) \cap \mathbb{H}^n$, $\psi(\overline{U}) = \overline{B_r(0)} \cap \mathbb{H}^n$ and $\psi(U') = B_{r'}(0) \cap \mathbb{H}^n$.

Now pick an open neighbourhood V of $0 \in \partial\mathbb{H}^n$ and $l > 0$ s.t. $\overline{V \times [0,l]} \subseteq B_r(0)$.

Considering $\overline{V}$ and V as submanifolds of $\mathbb{H}^n$ and $\psi^{-1}(\overline{V}) = \overline{\psi^{-1}(V)}$ and $\psi^{-1}(V)$ as submanifolds. Then

$$\psi|_{\overline{\psi^{-1}(V)}} : \overline{\psi^{-1}(V)} \to \overline{V}$$

and

$$\psi|_{\psi^{-1}(V)} : \psi^{-1}(V) \to V$$

are $\mathcal{C}^k$-diffeomorphisms. So it suffices to show that $\psi^{-1}|_{\overline{V} \times [0,l]} : \overline{V} \times [0,l] \to M$ fulfils the conditions given in Definition 1.2.4.

I think this should work... I think it would be nicer to do this directly in Lean and then later write up the maths proof. □

which will then allow us to derive

Theorem 1.2.15 (Collar neighbourhood theorem)**.** *Let M be a C^k-manifold with boundary, then ∂M is collared.*

2 The proof of the collar neighbourhood theorem for smooth manifolds

2.1 Prerequisites

2.1.1 Vector fields

Lemma 2.1.1. *Let M be a smooth n-manifold with boundary. Let B be a coordinate half ball with corresponding map $\varphi\colon B \to \mathbb{H}^n$. Then $\frac{\partial}{\partial x^n}$ is a smooth vector field on B which is inward-pointing when restricted to $\partial M \cap B$.*

Proof. The vector field is smooth by Example 8.2 in [Lee06]. Let $p \in B$. We need to show that $\left.\frac{\partial}{\partial x^n}\right|_p$ is inward-pointing. By Proposition 5.42 in [Lee06], it suffices to show that the n-th component of $\left.\frac{\partial}{\partial x^n}\right|_p$ is positive. Its n-th component is 1 so we are done. □

Lemma 2.1.2. *Let M be a smooth manifold, A a closed and U an open subset of M with $A \subseteq U$. Let $X\colon A \to TM$ be a smooth vector field along A. Then there is a smooth vector field $\tilde{X}\colon M \to TM$ with support in U which agrees with X on A.*

Proof. We first notice that the proof of Lemma 2.26 (Extension Lemma for Smooth Functions) in [Lee06] also works for the domain TM . Now we need to verify that a smooth map of the form

General Lemma?

$$\tilde{X} = \sum_{p \in A} \tau_p(x) \tilde{X}_p(x)$$

is a vector field. Here the $\tau_p x$ are bump functions and the $\tilde{X}_p$ local extensions of the vector field. Now locally this sum is finite. Module over $C^\infty(M)$ ... blah blah blah □

Lemma 2.1.3 (Problem 8-4 in [Lee06])**.** *Let M be a smooth n-manifold with boundary. There exists a smooth vector field on M, which is inward-pointing when restricted to the boundary.*

Proof. Cover ∂M with a countable number of regular coordinate half-balls $(B_i)_i$. This is possible by Exercise 1.42 in [Lee06]. Let us call the corresponding larger coordinate half-balls B'_i and the corresponding smooth coordinate map $\varphi_i\colon B'_i \to \mathbb{H}^n$. By Lemma 2.1.1, $\frac{\partial}{\partial x^n}$ is a smooth vector field on B'_i which is inward-pointing when restricted to $\partial M \cap B$. In particular, $\frac{\partial}{\partial x^n}$ is a smooth vector field along B_i. Using Lemma 2.1.2, we can extend these to smooth vector fields $X_i\colon M \to TM$ with support in B'_i and which agree with $\frac{\partial}{\partial x^n}$ on B_i. This is again inward-pointing which is easy to see using Proposition 5.42 in [Lee06] . Now take a partition of unity subordinate to the cover $\{B_i \mid i\} \cup M \setminus \partial M$. Then do the same thing as in the proof of Lemma 2.1.2 and get a global smooth vector field that is inward-pointing on the boundary. □

write Lemma

2.1.2 Flows and Flowouts

Lemma 2.1.4. *Let U be an open subset of $\mathbb{H}^n$ that has non-zero intersection with the boundary. Let V be a smooth vector field on U. Then there exists a smooth map $\delta_U \colon U \to \mathbb{R}+$ and a smooth open embedding Φ_U:*

Lemma 2.1.5. *Let M be a smooth n-manifold with boundary. Let N be a smooth vector field on M that is inward-pointing when restricted to the boundary. Let (U, φ) be a boundary chart. Then there exists a smooth map $\delta_U \colon U \to \mathbb{R}^+$ and a smooth open embedding $\Phi_U \colon P_U \to M$ where $P_U =:= \{(t,p) \in \mathbb{R} \times U \mid 0 \le t < \delta_U(p)\}$ ...*

How? -> Lemma

show $t \geq 0$

Proof. Consider the pushforward $\varphi_* X$ of X by φ on $\varphi(U) \subseteq \mathbb{H}^n$. Extend $\varphi_* X$ to a vector field V on an open subset U' of $\mathbb{R}^n$ with $U' \cap \mathbb{H}^n = U$ such that still $V^n(p) > 0$ for all $p \in \partial\mathbb{H}^n$. Now let $\theta_V \colon \mathcal{D} \to U'$ be the flow of V as given by the Fundamental Theorem of Flows (Theorem 9.12 in [Lee06]) where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain. Since $\varphi(U) \cap \partial\mathbb{H}^n$ is an embedded $n-1$-dimensional submanifold of U' and V is clearly nowhere tangent to it, we can apply the Flowout Theorem (Theorem 9.20 in [Lee06]) to get a smooth function $\delta'_U \colon \varphi(U) \cap \partial\mathbb{H}^n \to \mathbb{R}^+$ such that $\theta_V|_{\mathcal{O}} \colon \mathcal{O} \to U$ is a smooth embedding where $\mathcal{O} := \{(t,p) \in (\mathbb{R} \times (\varphi(U) \cap \partial\mathbb{H}^n) \cap \mathcal{D}) \mid 0 \le < \delta'_U(p)\}$.

Now define □

Lemma 2.1.6 (Boundary Flowout Theorem, Theorem 9.24 and Problem 9-11 in [Lee06])**.** *Let M be a smooth manifold with non-empty boundary. Let N be a smooth vector field on M that is inward-pointing when restricted to the boundary. Then there exists a smooth map $\delta \colon \partial M \to \mathbb{R}^+$ and a smooth embedding $\Phi_\delta \colon P_\delta \to M$ where $P_\delta := \{(t,p) \in \mathbb{R} \times \partial M \mid 0 \le t < \delta(p)\}$ such that $\Phi_\delta(P_\delta)$ is a neighbourhood of ∂M and for every $p \in \delta M$, $t \mapsto \Phi_\delta(t,p)$ is an integral curve of N starting at p.*

Proof. Cover ∂M with a countable number of boundary charts (U, φ). □

3 Inward Pointing Vectors in Manifolds with Corners

Defining the notion of inward pointing vectors in manifolds with corners is substantially harder than in manifolds with boundary without corners. Let us first examine why this is.

Here is a common definition for manifolds with boundary without corners:

Definition 3.0.1 ([Lee06] p. 136)**.** Let M be a smooth manifold with boundary without corners. Let $p \in M$. Then $v \in T_pM \setminus T_p\partial M$ is inward pointing if and only if there is an $\varepsilon > 0$ and a smooth curve $\gamma\colon [0, \varepsilon) \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

This definition cannot be used for manifolds with corners as in this case ∂M is not always a manifold and thus $T_p\partial M$ is not defined.

The existence of the $\gamma\colon\ : [0, \varepsilon) \to M$ by itself characterises whether a vector is inward pointing or tangent to the boundary, a notion which we will call *realisable*.

So for manifolds with corners we either need a completely new definition that intrinsically excludes the vectors tangent to the boundary or we need a definition of realisability and a way to exclude the tangent vectors.

3.1 Definitions intrinsically excluding vectors tangent to boundary

We first tried the first option and considered these candidates for definitions which we either came up with ourselves or found in the literature:

Definition 3.1.1 (Our idea inspired by Definition 3.0.1)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if there is an $\varepsilon > 0$ and a smooth curve $\gamma\colon [0, \varepsilon) \to M$ such that $\gamma(0) = p$, $\gamma'(0) = v$ and $\gamma((0, \varepsilon)) \subseteq \mathrm{int}(M)$.

Unfortunately, this definition doesn't exclude vectors tangent to the boundary:

Example 3.1.2. Take M to be the upper half plane $\mathcal{H}$ in $\mathbb{R}^2$ let p be $(0, 0)$ and let $\gamma\colon [0, 1) \to \mathcal{H}, x \mapsto x^2$. Then $\gamma'(0) = (1, 0)$ and thus by this definition $(1, 0)$ would be inward pointing even though it is clearly tangent to the boundary. See Figure 3.1 for an illustration of this example.

Definition 3.1.3 ([Mel96] Definition 1.13.10)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if $\mathrm{D}_v f(p) \geq 0$ for every non-negative $f \in C^\infty(M)$ that is zero on ∂M.

For some manifolds this definition includes vectors tangent to the boundary:

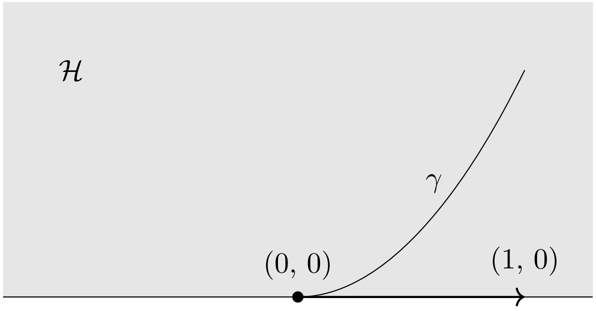

Figure 3.1: Illustration of Example 3.1.2

Example 3.1.4. Take M to be the upper half plane $\mathcal{H}$ in $\mathbb{R}^2$ and let p be $(0,0)$. Let f be any non-negative function in $C^\infty(\mathcal{H})$ that is zero on $\partial\mathcal{H}$. Then $\mathrm{D}_{(0,1)}f(0,0) = 0$. Thus $(0,1)$ is inward pointing by this definition even though it is clearly tangent to the boundary.

To remedy this we came up with this definition:

Definition 3.1.5 (Our idea inspired by Definition 3.1.3)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if there is a non-negative $f \in C^\infty(M)$ that is zero on ∂M such that $\mathrm{D}_v f(p) > 0$.

Unfortunately, both of the above definitions do not work for manifolds with corners:

Example 3.1.6. Take M to be the right upper quadrant Q_I and let p be $(0,0)$. Let f be any non-negative function in $C^\infty(Q_I)$ that is zero on $\partial\mathcal{Q}_\mathcal{I}$. Then clearly $\mathrm{D}_{(0,1)}f(0,0) = \mathrm{D}_{(1,0)}f(0,0) = 0$. This means that the Jacobian is zero and since f is smooth and thus differentiable at $(0,0)$, this means that $\mathrm{D}_v f(0,0) = 0$ for all vectors v.

So in the case of Definition 3.1.3, this would mean that any vector was inward pointing, and in the case of Definition 3.1.5, no vector would be inward pointing.

Lastly, here is another popular definition that falls into this category:

Definition 3.1.7 (E.g. [Joy09] Definition 2.2)**.** Let M be a smooth manifold with corners. Let $p \in M$ be a corner point of codimension k, i.e. let there be a chart (U,φ) with $U \subseteq \mathbb{R}^n_k = [0,\infty)^k \times \mathbb{R}^{n-k}$ such that $0 \in U$ and $\varphi(U) = p$. Then a vector $v \in T_pM$ is inward pointing if and only if it is in $\mathrm{d}\varphi_0((0,\infty)^k \times \mathbb{R}^{n-k})$ where $\mathrm{d}\varphi_0 \colon \mathbb{R}^n = T_0\mathbb{R}^n_k \to T_pM$.

This definition is of course correct, however it depends on a choice of chart. In Mathlib definitions are generally preferred to not depend on choices.

Why exactly? Reference?

3.2 Definitions through realisability

After failing to find a usable definition intrinsically excluding vectors, we moved on to defining realisable vectors. Where as mentioned above, we want a realisable vector to be one that is either inward pointing or tangent to the boundary.

Firstly, we could only find the following way to define inward pointing vectors from realisable ones:

Definition 3.2.1 ([Amr21])**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if it is in the interior of the set of realisable vectors.

Now we can move on to finding a definition of realisability.
We first note conditions for such a definition:

Remark 3.2.2. A sensible definition of realisability needs to fulfil these two conditions:

(i) The definition needs to be preserved by local diffeomorphisms (in particular it should be preserved by smooth charts).

(ii) The definition should reduce to a sensible notion on the model space that is easily verifiable in the cases of the typical examples (e.g. upper half space in $\mathbb{R}^n$).

Let us first discuss what this "sensible notion" in Remark (ii) should be.
We first need the following definition:

Definition 3.2.3 (Contingent Tangent Cone, [AF90] after Definition 4.1.1)**.** Let E be a normed vector space, $K \subset E$ and $x \in \overline{E}$. Then the *contingent tangent cone* to K at x is the set of all $v \in E$ such that there is a sequence of scalars $c \colon \mathbb{N} \to \mathbb{R}_{\geq}$ and a sequence of vectors $d \colon \mathbb{N} \to E$ such that

(i) $\lim_{n\to\infty} d_n = 0$,

(ii) $x + d_n \in K$ for all $n \in \mathbb{N}$, and

(iii) $\lim_{n\to\infty} c_n \cdot d_n = v$.

We denote this tangent cone by $T_K(x)$.

In `Mathlib` this is referred to as `posTangentConeAt`. In the following we will only consider contingent tangent cones to convex sets for which the contingent tangent cone agrees with other notions of tangent cones such as the Clarke tangent cone ([AF90] Proposition 4.2.1). Thus we will refer to it simply as the tangent cone.

This captures exactly our notion of realisable: intuitively, the tangent cone $T_K(x)$ is composed of all vectors at x that are either inward pointing or tangent to K. You can see examples of tangent cones in Figure 3.2.

So we can take the "sensible notion" in Remark (ii) to mean that a vector v should be inward pointing to a point x in the model space if and only if v is in the tangent cone at the point x to the model space.

Now let us consider possibilities for the definition of realisability. The first was already provided to us in Definition 3.0.1:

Definition 3.2.4 ([Lee06] p. 136)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is *realisable* if and only if there is an $\varepsilon > 0$ and a smooth curve $\gamma \colon [0, \varepsilon) \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

It is easy to show that this definition fulfils Remark (i). It also fulfils Remark (ii) but this is surprisingly hard to show for general model spaces H. Taking a vector tangent to the boundary, it is hard to describe a method for picking a smooth curve $\gamma \colon [0, \varepsilon) \to H$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

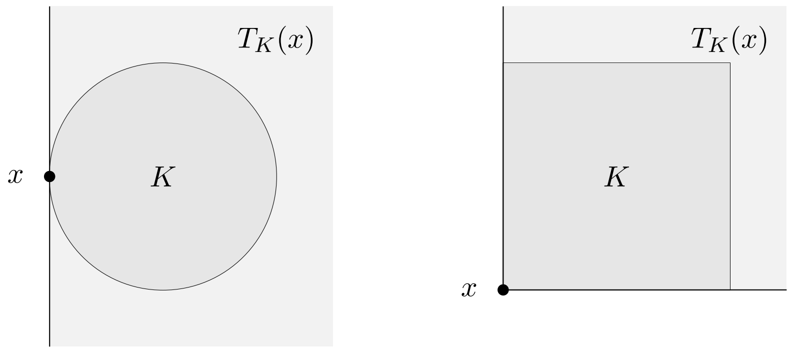

Figure 3.2: Two examples of tangent cones

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